PROBABILITY, STATISTICS AND COMPUTATIONAL MATHEMATICS

MODULES 4 AND 5

High-yield notes, formulas, worked examples, and question bank

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Exam Focus

Scope. These notes cover only Module 4 (Mathematical Logic) and Module 5 (Counting and Mathematical Induction), matching the supplied module PDFs. The exam timetable lists the MCA101 paper for Friday, 30 October 2026, 9:30–10:30 a.m.; the marks pattern was not supplied, so the question lengths below are revision guidance rather than a confirmed blueprint. For proof and calculation questions, show the rule or formula, substitute values, and state the conclusion clearly.

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Module 4: Mathematical Logic

4.1 Propositions and connectives

A **proposition** is a declarative statement with exactly one truth value: true (T) or false (F). Questions, commands, and open sentences containing an unspecified variable are not propositions.

Connective	Notation	Meaning	Truth condition
Negation	$\neg p$	not p	True exactly when p is false
Conjunction	$p \wedge q$	p and q	True when both are true
Inclusive OR	$p \vee q$	p or q or both	False only when both are false
Exclusive OR	$p \oplus q$	exactly one of p, q	True when their values differ
Implication	$p \to q$	if p, then q	False only for $p = T, q = F$
Biconditional	$p \leftrightarrow q$	p iff q	True when their values agree

For n variables, a complete truth table has 2^n rows. The implication $p \to q$ is logically equivalent to $\neg p \vee q$, and $p \leftrightarrow q$ is equivalent to $(p \to q) \wedge (q \to p)$.

p	q	$\neg p$	$p \wedge q$	$p \vee q$	$p \oplus q$	$p \to q$	$p \leftrightarrow q$
T	T	F	T	T	F	T	T
T	F	F	F	T	T	F	F
F	T	T	F	T	T	T	F
F	F	T	F	F	F	T	T

4.2 Tautology, contradiction, contingency

- A **tautology** is true on every row (example: $p \vee \neg p$).
- A **contradiction** is false on every row (example: $p \wedge \neg p$).
- A **contingency** is true on some rows and false on others (example: $p \wedge q$).

To classify a formula, construct its full truth table and inspect the final column.

4.3 Logical equivalences

Two formulas are logically equivalent, written $P \equiv Q$, when their truth values agree for every assignment. Use a truth table or the following laws.

Law	Form 1	Form 2
Identity	$p \wedge T$	p; $p \vee F = p$
Domination	$p \vee T$	T; $p \wedge F = F$
Idempotent	$p \vee p$	p; $p \wedge p = p$
Double negation	$\neg\neg p$	p
Commutative	$p \vee q$	$q \vee p$; similarly for $\wedge$
Associative	$(p \vee q) \vee r$	$p \vee (q \vee r)$; similarly for $\wedge$
Distributive	$p \wedge (q \vee r)$	$(p \wedge q) \vee (p \wedge r)$
De Morgan	$\neg(p \wedge q)$	$\neg p \vee \neg q$
De Morgan	$\neg(p \vee q)$	$\neg p \wedge \neg q$
Absorption	$p \vee (p \wedge q)$	p; dual: $p \wedge (p \vee q) = p$
Implication	$p \to q$	$\neg p \vee q$
Biconditional	$p \leftrightarrow q$	$(p \to q) \wedge (q \to p)$

Converse, inverse, contrapositive

For $p \to q$: the **converse** is $q \to p$; the **inverse** is $\neg p \to \neg q$; the **contrapositive** is $\neg q \to \neg p$. The original and contrapositive are equivalent. The converse and inverse are equivalent to each other, but are not generally equivalent to the original.

Do Not Confuse

For "if p, then q," do not swap the hypothesis and conclusion and assume the result is equivalent. Only the contrapositive is guaranteed equivalent.

4.4 Predicates and quantifiers

A **predicate** $P(x)$ is a statement whose truth depends on the value assigned to x. A domain specifies the allowed values. Quantifiers turn predicates into propositions:

- $\forall x\, P(x)$: for every x in the domain, $P(x)$ is true.
- $\exists x\, P(x)$: there is at least one x in the domain for which $P(x)$ is true.

Negation rules: $\neg\forall x\, P(x) \equiv \exists x\, \neg P(x)$ and $\neg\exists x\, P(x) \equiv \forall x\, \neg P(x)$. For example, "not everyone passed" means "someone did not pass," not "nobody passed."

Nested quantifiers

Quantifier order can change meaning. Over a suitable domain, $\forall x \exists y\, R(x, y)$ says each x may have its own y; $\exists y \forall x\, R(x, y)$ says one fixed y works for every x. In general, these are not equivalent. To disprove a universal statement, give a counterexample; to prove an existential statement, give a witness.

4.5 Rules of inference

An argument is valid when its conclusion follows from its premises. Common valid forms:

Rule	Premises	Conclusion
Modus ponens	$p \to q$, p	q
Modus tollens	$p \to q$, $\neg q$	$\neg p$
Hypothetical syllogism	$p \to q$, $q \to r$	$p \to r$
Disjunctive syllogism	$p \lor q$, $\neg p$	q
Addition	p	$p \lor q$
Simplification	$p \land q$	p (or q)
Conjunction	p, q	$p \land q$
Resolution	$p \lor q$, $\neg p \lor r$	$q \lor r$

4.6 Normal forms

An **elementary product** (minterm) is an AND of literals; an **elementary sum** (maxterm) is an OR of literals. A literal is a variable or its negation.

- **DNF**: OR of AND terms, e.g. $(p \land q) \lor (\neg p \land r)$.
- **CNF**: AND of OR clauses, e.g. $(p \lor q) \land (\neg p \lor r)$.

To convert algebraically, first remove $\to$ and $\leftrightarrow$, push negations inward using De Morgan's laws, then distribute. For canonical forms, use truth-table rows: DNF takes the rows where the formula is true (minterms); CNF takes rows where it is false (maxterms).

Exam Focus

Worked conversion. $p \to q \equiv \neg p \lor q$, which is already CNF. To write it as DNF, expand with $x \equiv (x \land T)$ and distribute, or use the true rows: $(\neg p \land \neg q) \lor (\neg p \land q) \lor (p \land q)$.

4.7 Exam answer patterns

Answer Skeleton

Truth table / tautology: list variables; make all 2^n assignments; add intermediate connective columns; evaluate the target column; classify and state the result.
Equivalence proof: start with one side; name each identity used; transform step by step to

the other side.
Inference: number premises; identify the rule applied at each line; show the conclusion as the last line.
Quantifier negation: negate the quantifier, switch $\forall \leftrightarrow \exists$, then negate the predicate.

Module 5: Counting and Mathematical Induction

5.1 Addition and multiplication principles

Addition rule: if a choice can be made in one of mutually exclusive cases with $a_1, a_2, \ldots, a_k$ outcomes, total outcomes are $a_1 + a_2 + \cdots + a_k$. If cases overlap, remove duplicate counts with inclusion–exclusion.

Multiplication rule: if a process has successive choices with $a_1, a_2, \ldots, a_k$ possibilities at each stage, total outcomes are $a_1 a_2 \cdots a_k$. The number may vary by stage; multiply the actual choices available at each step.

Exam Focus

Examples. Three-letter strings followed by three digits, repetition allowed: $26^3 \cdot 10^3$. Select one boy from 28 and one girl from 18: $28 \cdot 18 = 504$. A die rolled five times: $6^5 = 7776$.

5.2 Factorials and permutations

$n! = n(n-1)\cdots 2 \cdot 1$, and $0! = 1$. A permutation is an ordered arrangement; changing order creates a different result.

$$ {}^nP_r = P(n,r) = \frac{n!}{(n-r)!} \quad \text{(choose and arrange } r \text{ distinct objects from } n\text{)}. $$

If repetition is allowed and each of r positions has n choices, there are n^r strings. If all n objects are arranged with repeated indistinguishable types of multiplicities $n_1, \ldots, n_k$, the number is $\frac{n!}{n_1! \cdots n_k!}$.

Exam Focus

Worked examples. Four distinct letters selected and ordered from six: $6P4 = 6!/2! = 360$. Arrangements of ROOTS: $5!/2! = 60$ (two O's). Six-digit PIN with digits 0–9 and repetition: 10^6; leading zero is allowed for a PIN.

5.3 Combinations

A combination is a selection where order does not matter:

$$ \binom{n}{r} = C(n,r) = \frac{n!}{r!(n-r)!}, \qquad nPr = \binom{n}{r} r!. $$

Useful identities: $\binom{n}{0} = \binom{n}{n} = 1$; $\binom{n}{1} = n$; $\binom{n}{r} = \binom{n}{n-r}$; $\binom{n}{r} + \binom{n}{r-1} = \binom{n+1}{r}$; $\sum_{r=0}^{n} \binom{n}{r} = 2^n$.

Exam Focus

Worked examples. Teams of 3 from 8: $\binom{8}{3} = 56$. A committee of 3 from 6 faculty including both named professors: choose the third member from the other four, so 4. From 5 boys and 4 girls, 3-person committees with at least one girl: $\binom{9}{3} - \binom{5}{3} = 84 - 10 = 74$.

Do Not Confuse

Use permutations when order or role assignment matters; use combinations when only membership matters. A PIN permits leading zero; a number usually does not. "At least one" is often quickest by counting the total and subtracting the zero-case.

5.4 Pigeonhole principle

Basic principle: placing N objects into k boxes with $N > k$ guarantees a box with at least two objects. **Generalized principle:** some box contains at least $\lceil N/k \rceil$ objects. To guarantee at least r objects in one of k boxes, the minimum required is

$$k(r-1)+1.$$

Example: 47 students in 8 groups gives a group of at least $\lceil 47/8 \rceil = 6$. To guarantee 5 birthdays in the same month, $12(5-1)+1 = 49$ people suffice.

5.5 Inclusion–exclusion principle

For two finite sets:

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

For three sets:

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.$$

For exactly two sets among three, add the three pairwise intersections and subtract three times the triple intersection. For exactly one, add the single-set totals, subtract twice the pairwise intersections, then add three times the triple intersection.

Exam Focus

> **Worked examples.** Of 80 students, 50 use Android and 45 iPhone, and everyone uses at least one: both $= 50 + 45 - 80 = 15$. For the supplied language survey, at least one $= 50 + 45 + 40 - 20 - 18 - 15 + 8 = 90$.

5.6 Mathematical induction

To prove $P(n)$ for all integers $n \geq n_0$:

1. State $P(n)$ and the starting value n_0.
2. **Base case:** verify $P(n_0)$ directly.
3. **Inductive hypothesis:** assume $P(k)$ is true for an arbitrary $k \geq n_0$.
4. **Inductive step:** use that assumption to prove $P(k+1)$.
5. Conclude by induction that $P(n)$ holds for every integer $n \geq n_0$.

Exam Focus

> **Worked proof.** Prove $1 + 2 + \cdots + n = \frac{n(n+1)}{2}$ for $n \geq 1$. Base: $n = 1$ gives $1 = 1$. Assume $1 + \cdots + k = \frac{k(k+1)}{2}$. Then $1 + \cdots + k + (k+1) = \frac{k(k+1)}{2} + (k+1) = \frac{(k+1)(k+2)}{2}$. This is the formula for $k+1$; hence it holds for all $n \geq 1$.

Do Not Confuse

> In the inductive step, assume the claim only for k, then derive the $k+1$ case. Do not assume the statement you are trying to prove for $k+1$. Include both the base case and the final conclusion.

5.7 Quick method selector

Clue in question	Method
"either ... or" in disjoint cases	Addition rule
several stages / "and then"	Multiplication rule
arrange, order, rank, positions	Permutation
choose, select, committee, team	Combination
guarantee a repeated category	Pigeonhole principle
overlapping groups / at least one set	Inclusion–exclusion
prove for every natural number	Mathematical induction

Important Exam Topics

- Truth tables; tautology, contradiction, contingency; propositional equivalence laws.
- Converse, inverse, contrapositive; translating statements; predicates, quantifiers, and quantifier negation.
- Rules of inference and DNF/CNF conversion.
- Addition and multiplication rules; factorials; permutations with/without repetition; combinations and committee constraints.
- Generalized pigeonhole principle; two- and three-set inclusion–exclusion.
- Mathematical induction proofs, especially sums and divisibility.

Answer Templates

Prove a logical equivalence [5–10 marks]

Answer Skeleton

Write the expression; remove implications; apply the named equivalence law one step at a time; reach the target expression. Alternatively, show identical truth-table columns. Name each law to make the reasoning clear.

Solve a counting problem [5–10 marks]

Answer Skeleton

Identify whether order matters and whether repetition is allowed; define the cases or stages; state the counting rule/formula; substitute values; simplify; answer in context. For restrictions, count valid cases only or use complement counting.

Prove by induction [5–10 marks]

Answer Skeleton

Let $P(n)$ be the claim. Verify the base case. Assume $P(k)$. Starting from the left side for $k+1$, use the inductive hypothesis and simplify to the required right side. State that the claim follows for all specified n.

Question Bank

Definitions / short answers

1. Define proposition, predicate, tautology, contradiction, and contingency.

2. State the truth condition for implication and biconditional.
3. State De Morgan's laws and the negation rules for quantifiers.
4. Define DNF and CNF, with one example of each.
5. Define permutation and combination; state the relation between them.
6. State the generalized pigeonhole principle and two-set inclusion–exclusion formula.

Practice problems

1. Construct truth tables for $p \vee \neg q$ and $(p \vee q) \vee \neg p$; classify each.
2. Use equivalence laws to show $p \to (q \to p)$ is a tautology.
3. Find the converse, inverse, and contrapositive of "If a number is divisible by 4, then it is even."
4. Negate: "Every student solved some problem." Express the result with quantifiers.
5. Identify the inference rule: $p \to q$, $q \to r$, therefore $p \to r$.
6. Convert $p \to (q \wedge r)$ to CNF and DNF.
7. How many 4-digit numbers can be made from digits 1–9 without repetition?
8. How many arrangements of INDEPENDENCE are there? How many start with P? (Use letter multiplicities carefully.)
9. From 6 men and 5 women, how many 4-person committees include at least one woman?
10. How many integers from 1 to 100 are divisible by 2 or 5?
11. What minimum number of people guarantees that at least 4 were born on the same day of the week?
12. In a class, 35 study Statistics, 28 study Discrete Mathematics, and 12 study both. Find at least one, only Statistics, only Discrete Mathematics, and exactly one.
13. Prove by induction that $1 + 3 + 5 + \cdots + (2n - 1) = n^2$ for $n \geq 1$.
14. Prove by induction that $3 \mid (n^3 + 2n)$ for every positive integer n.

Last-Minute Revision Page

LOGIC

$p \to q \equiv \neg p \vee q$
$p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$
$\neg \forall x P \equiv \exists x \neg P$
$\neg \exists x P \equiv \forall x \neg P$

COUNTING

Addition: sum disjoint cases
Multiplication: product successive choices
$nPr = n!/(n - r)!$
$nCr = n!/[r!(n - r)!]$
Repetition: n^r
Identical objects: $n!/\prod n_i!$

GUARANTEES / OVERLAP

Pigeonhole: at least $\lceil N/k \rceil$
Guarantee r: $k(r - 1) + 1$
$|A \cup B| = |A| + |B| - |A \cap B|$
Three-set pattern: singles − pairs + triple

INDUCTION

Base case → assume $P(k)$ → prove $P(k + 1)$ → conclude

DO NOT CONFUSE

Order: permutation; no order: combination.
Contrapositive equivalent to original; converse generally not.
At least one: consider complement.

Source Scope

Prepared from the supplied `Module 4 Part 1.pdf` and `Module 5.pdf` in the Statistics folder, following the structure and visual style of `notes-template.tex`. Module 4's listed syllabus also includes predicates/quantifiers, nested quantifiers, and rules of inference; concise standard definitions and rules are included to cover those listed topics where the slides provide limited detail. Verify any instructor-specific conventions against class notes.

Study principle: identify the rule, show the working, and state the conclusion.